

Designing Human–Computer Integration in an Exertion Context

Josh Andres

Exertion Games Lab, Monash University, AUS
ja@exertiongameslab.org

ABSTRACT

Human–computer interaction researchers continue to support the exerting user to promote the many benefits of being physically active. Recently, due to technological advances, systems emerged that can continuously sense, interpret and automatically act on information, opening the opportunity to design human–computer "integration", where the user and the system work in a partnership. However, today's knowledge gap is how to design human–computer integration experiences in an exertion context. In this talk, I present how I aimed to close this gap through the design of three eBike systems that "act on" different data types (motion, traffic light data, and EEG (electroencephalography)) to explore integrations with the exerting body. The investigation informed the creation of the "integrated exertion framework", which can guide interaction designers on how to, and in an inclusive manner, amplify a person's sensations of their abilities in an exertion context to create "superpower", "co-operative" and "symbiotic" human–computer partnerships.

CCS CONCEPTS

• **Human-centered Computing**; • **Human-computer interaction (HCI)**; • **Interaction paradigms**;

KEYWORDS

human–computer interaction, human–computer integration, exertion, whole body

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1 DESIGNING FOR INTEGRATED EXERTION

The emerging area of human–computer integration offers a wealth of opportunities for HCI practitioners, especially in an exertion context. Today, there is limited design knowledge on how to design interactive systems that offer integration with the exerting body. I describe this intersection as "integrated exertion", and more precisely, I define it as the intersection between human–computer integration, where the user and computer co-operate in partnership [6] and exertion support, where the user invests physical effort

[7, 8]. This is an emerging area in HCI with potential application across various domains.

Therefore, my work investigated the following question:

How do we design integrated exertion experiences?

I explored this question by designing and studying three eBike systems that offer different forms of integration with the exerting body by using different data types, from motion [2, 3], traffic light [4], to EEG (electroencephalography) [5]. The studies resulted in detailed themes and design tactics to explore the design space that integrated exertion offers, and this knowledge is synthesised in the "Integrated Exertion Framework" [1].

This work aims to support a vision of healthy engagement with technology by melding the exerting body and interactive systems through a non-invasive integration approach. This approach results in the following societal benefits:

- The work promotes the design of technology that offers physical activity benefits while enabling users to move freely without screens.
- The work responds to eBike cycling challenges (such as increased accidents with eBikes near traffic lights) by presenting potential cycling futures that more people could use and support environmental choices.
- The work frames the design of "integration experiences" using a "human values" perspective to signify inclusivity and accessibility of different bodies and neurodiverse qualities.
- The work guides interaction designers how to, in an ethical way, amplify a person's sensations of their abilities, almost as if they were "bionic".

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